# Sleep, Screen Time and Productivity Analysis

Final Project Report
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## Abstract

This project explores whether daily productivity is associated with sleep duration and screen time. The motivation comes from observing fluctuations in personal academic performance and questioning whether lifestyle factors influence productivity.

## Introduction

Productivity is affected by multiple behavioral factors. Sleep quality and duration are often linked with cognitive performance, while prolonged screen exposure may influence focus and efficiency. This study investigates these relationships using manually collected daily observations.

## Research Question

How do sleep hours and screen time relate to the number of tasks completed per day?

## Dataset Description

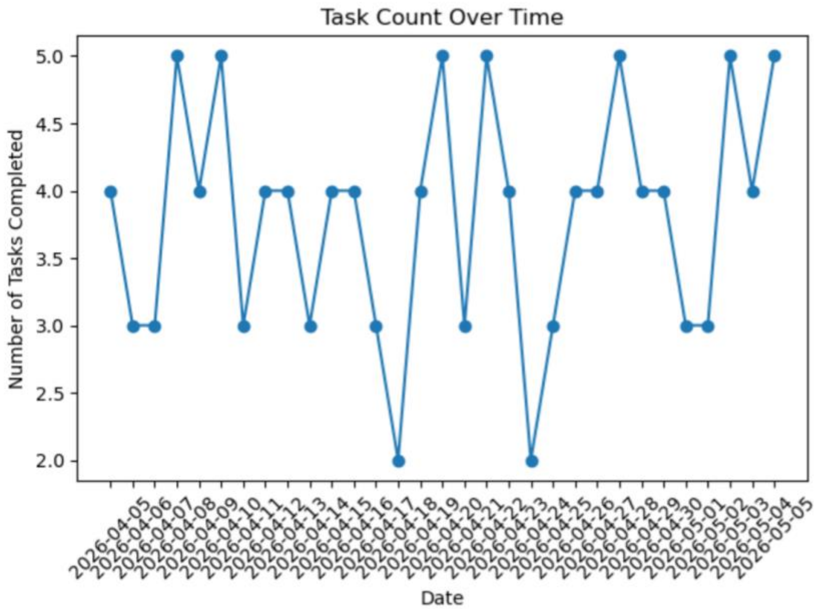
The dataset contains approximately 30 days of manually collected observations including date, sleep duration (hours), daily screen time (hours), and completed task count.

## Methodology

Exploratory Data Analysis (EDA), correlation analysis, scatter plots, and linear regression were used.

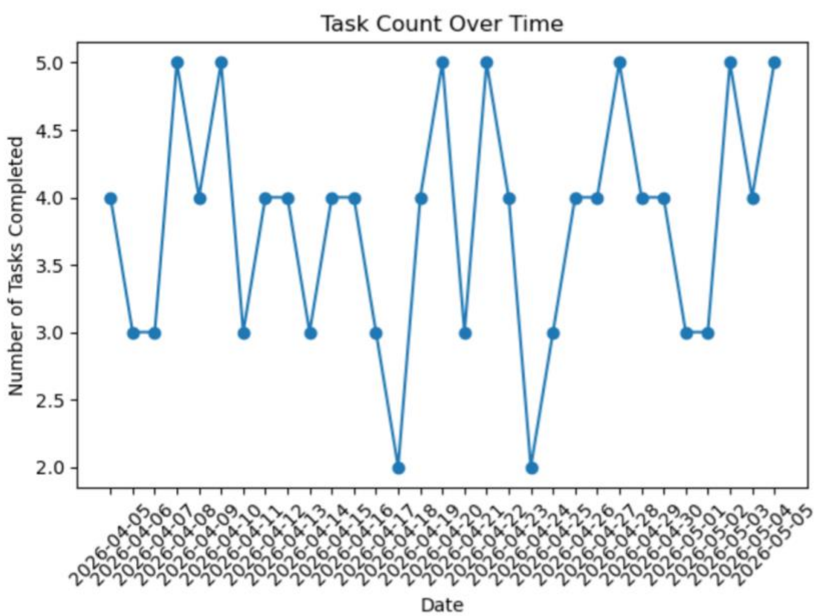
## Results and Visualizations

### Task Count Over Time



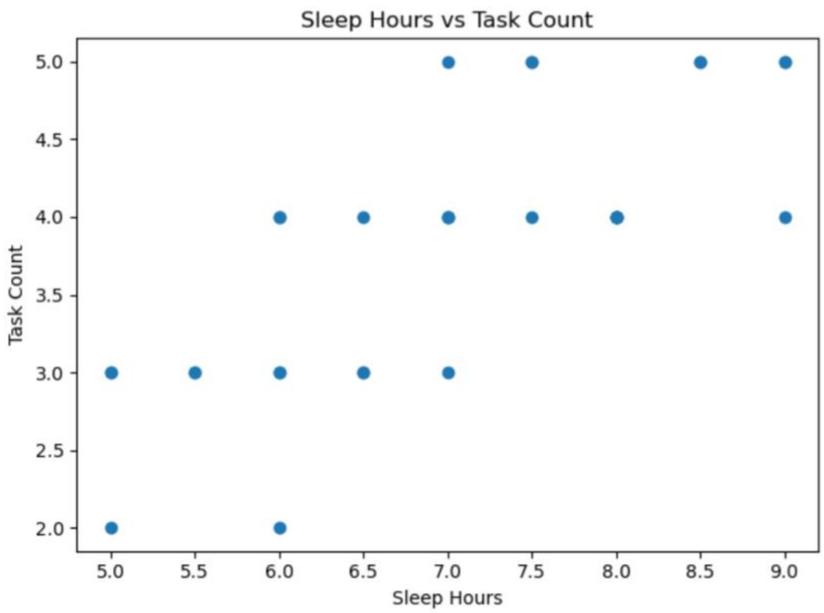

Productivity fluctuates between 2 and 5 tasks. Some periods show consistent productivity while occasional drops may reflect fatigue or workload variation.

### Sleep Hours vs Task Count



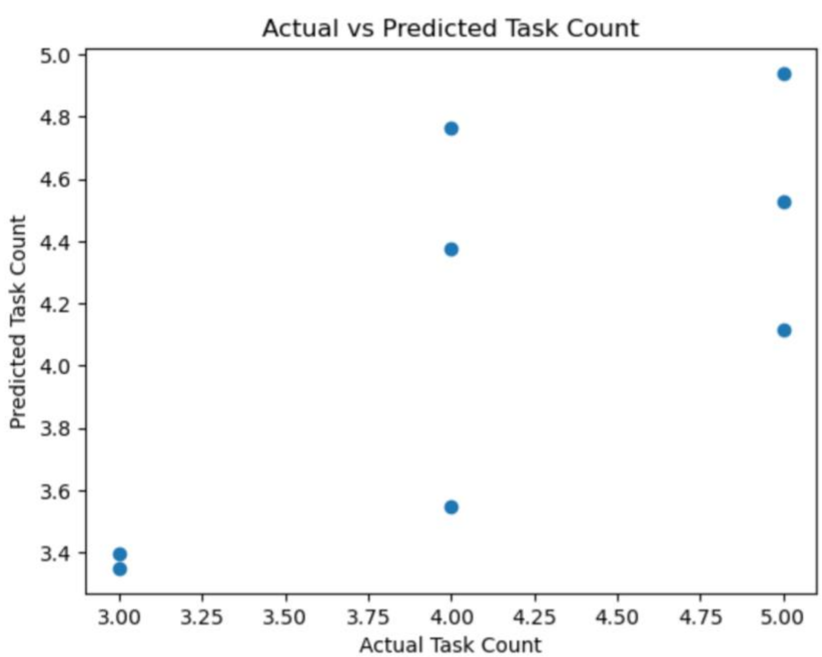

The scatter plot suggests a positive relationship: higher sleep duration is often associated with higher task completion.

## Screen Time vs Task Count




Screen time does not show a strong positive association and may negatively affect productivity.

## Actual vs Predicted Task Count




The regression model captures trends but prediction errors remain due to the small dataset.

## Correlation Analysis

Correlation values indicated a positive correlation between sleep hours and task count, while screen time correlated negatively with productivity. This supports previous findings emphasizing sleep as an important factor in cognitive performance.

## Machine Learning Analysis

A linear regression model was trained using sleep duration and screen time to predict daily task count. The model achieved an $R^2$ score around 0.54, indicating moderate predictive performance. Given the limited dataset size, results should be interpreted as exploratory rather than definitive.

## Limitations

The dataset is manually collected, relatively small, and based on one individual. External factors such as stress, workload, deadlines, exercise, and mood were not included.

## Future Work

Future studies could include longer observation periods, more participants, stress measurements, exercise habits, and advanced ML models.

## Conclusion

The findings suggest sleep has a stronger relationship with productivity than screen time. Improving sleep habits may contribute to improved daily performance.